

Atmospheric Turbulence-Induced Secret Key Distribution in FSO-based UAV Systems

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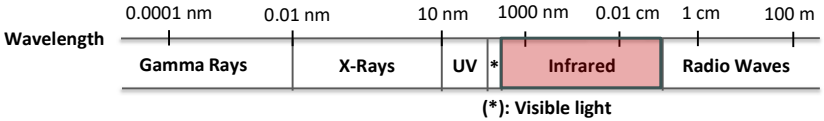
Aizu Wakamatsu, September, 2026

Outline of Presentation

I. Introduction
II. System Model
III. Numerical Results and Discussions

Free-Space Optics (FSO): Advantages and Security Challenges

- **FSO is a promising candidate for high-speed connections**
  - Using infrared bands → support extremely high data rates (~ Gbps, up to Tbps)
  - Low power consumption and License-free spectrum → suitable for UAV deployment



(*): Visible light

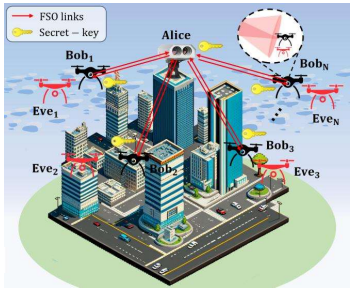
- **Security Challenges in FSO**
  - FSO links operate over open free-space

→ Optical signals can be intercepted by eavesdropper

Key question: How can we address security challenges in FSO?

⬇

One possible solution is Secure key establishment



Existing Approaches for Key Establishment in FSO Systems

- **Comparison of Existing Approaches**

Approaches	Security basis	Hardware complexity	Limitations
Quantum Key Distribution	Quantum physics	High	Requires quantum hardware
Public Key Cryptography	Computational hardness	Low	Quantum attack risk
Physical Layer Key Generation	Channel randomness	Low	Requires channel reciprocity

→ Physical Layer Key Generation: A lightweight solution (No specialized hardware required)

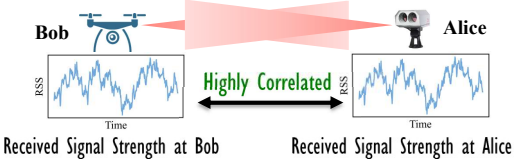


Fig. 1: Illustration of channel reciprocity between Bob and Alice

Physical Layer Key Generation

- **Raw key generation**

- Step 1. Channel Probing: Alice and Bob exchange pilot signals to observe the channel

- Step 2. Quantization: The measured values are converted into binary bits (0 and 1)

- **Post-processing**

- Step 3. Key Reconciliation: ECC is used to correct errors and achieve identical keys

- Step 4. Privacy Amplification: The key is refined to remove any information that eavesdropper might have

Question:
What happens to channel reciprocity when pointing errors occur?

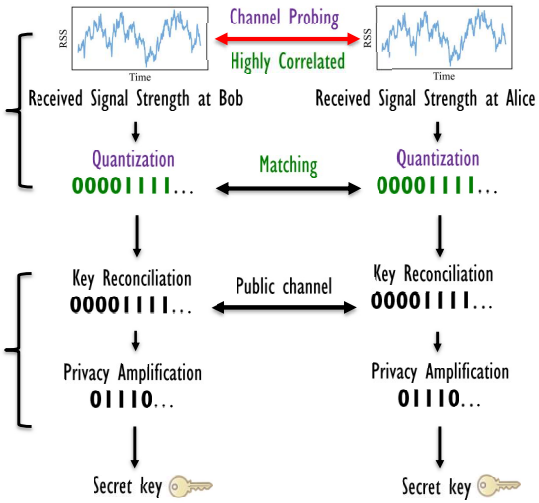


Fig.2: Key generation process under perfect channel reciprocity

Physical Layer Key Generation: Research Problems

❑ Problem 1: Pointing errors

- Pointing errors damage channel reciprocity → Key mismatch → Reduce secret key rate (SKR)

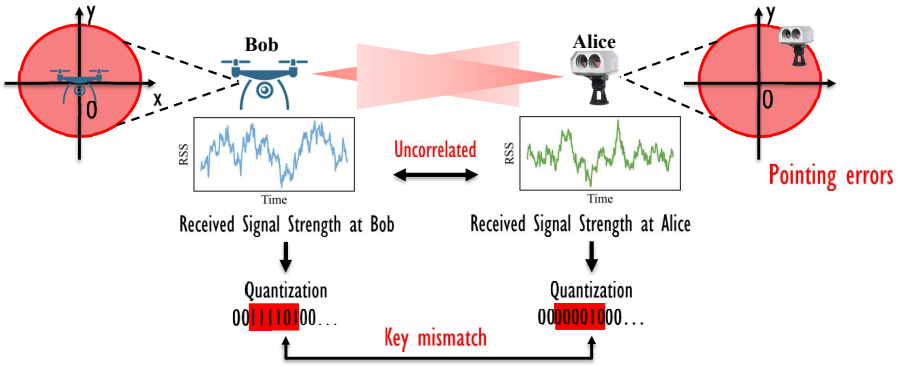


Fig.3: Key generation under imperfect channel reciprocity

❑ Problem 2: Low SKR

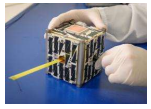
- The SKR $\sim \frac{1}{T_c}$; Where T_c: Channel coherent time

Physical Layer Key Generation: Pointing Error Solutions

❑ Conventional scheme : APT mechanisms

- APT is widely used in free-space optical (FSO) communication systems to establish and maintain accurate beam alignment.
- Limitations: APT systems often suffer from high complexity, power consumption, and payload requirements

→ Making them less suitable for lightweight UAVs and small satellites (CubeSats)

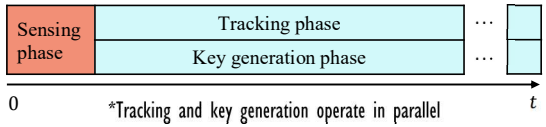


→ A CubeSat built measures about 10 centimeters on a side.

❑ Alternative solution: Sensing-assisted optical wireless communication

- Key idea: Time-division is used to allow sensing, tracking and key generation phase to share the same transceiver hardware, where the sensing phase and tracking phase acts as the function of an APT system

→ Reduce resource consumption

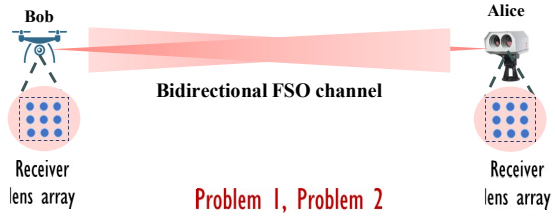


Physical Layer Key Generation: Low SKR solution

❑ Problem 2: Low SKR

The SKR $\sim \frac{1}{T_c}$; Where T_c: Channel coherent time

→ Solution: We propose using a lens array to increase the sources of randomness for key generation



Final goal:

(1) Design sensing and beam tracking system to mitigate pointing errors

(2) Improve the SKR by exploiting spatial diversity in a lens-array-based FSO system

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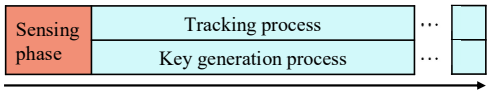
System Model

❑ System Model

- Bob: A hovering UAV (without APT system)
- Alice: Base station (with APT system)
- Alice and Bob generate secret keys from the estimated turbulence channel.

❑ Proposed Scheme

1. Sensing phase: UAV accurately estimates the receiver's location
2. Tracking phase: Alice send estimated pointing error to Bob for beam tracking controller
3. Key generation phase: Alice and Bob extract the randomness of the estimated turbulence channel to generate secret keys



*Tracking and key generation operate in parallel

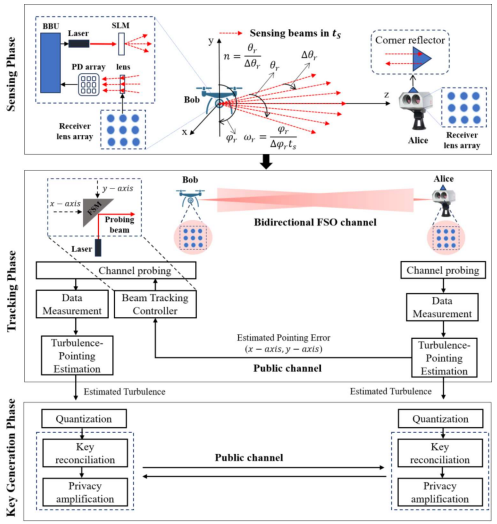


Fig.4: System Architecture

System Model – 1. Sensing Phase

❑ Working principle

- 2D Scanning: The UAV uses multiple narrow beams to scan in both horizontal (azimuth) and vertical (elevation) directions.
- Position Estimation: The UAV uses signals reflected from the corner reflector to locate the base station

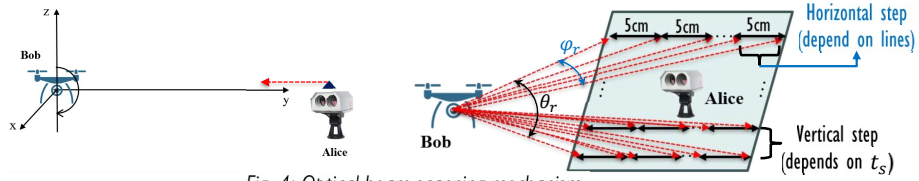


Fig. 4: Optical beam scanning mechanism

❑ Alignment Accuracy

→ Accuracy depends on Angular Scanning Resolution

- $Horizontal\ step = \frac{Horizontal\ scanning\ range(\varphi_r)}{Number\ of\ scanning\ lines}$
- $Vertical\ step = \frac{Vertical\ scanning\ range(\theta_r)}{Scanning\ frequency\ t_s}$

→ Longer sensing time (t_S) → Higher scanning resolution→ Lower radial misalignment

→ After sensing phase, the system still suffers from jitter

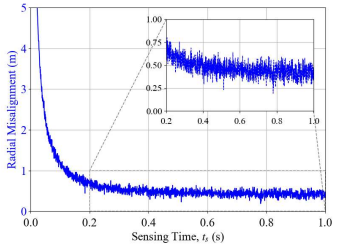
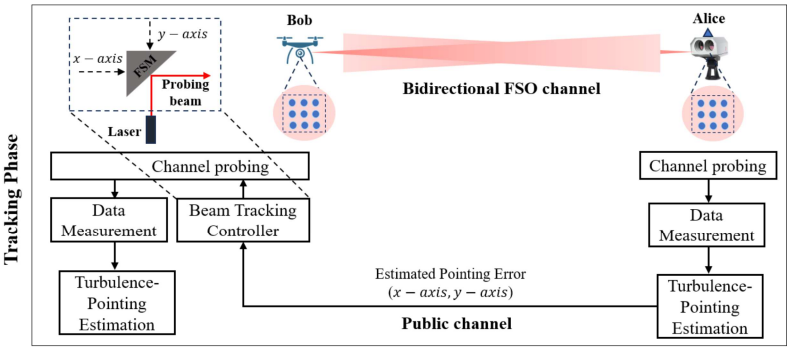


Fig. 5: Radial misalignment with sensing time

System Model – 2. Tracking phase



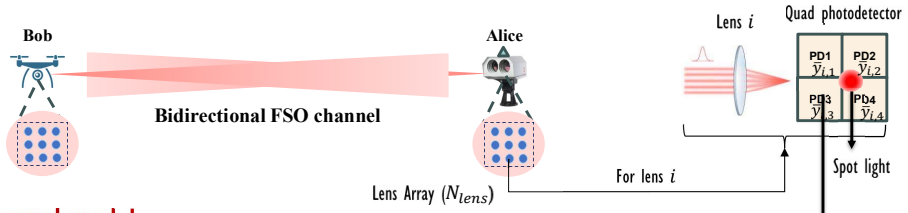
I&2. Channel Probing and Data Measurement: Alice and Bob exchange probing signals to estimate the channel characteristics.

3. Turbulence Pointing Estimation: Based on the received signals, atmospheric turbulence and pointing errors are estimated

4: Beam Tracking Controller: Based on the estimated pointing error, the beam tracking controller adjusts the beam direction to maintain accurate alignment.

2. Tracking phase – Channel probing and Data measurement

I&2. Channel Probing and Data Measurement: Alice and Bob exchange probing signals to estimate the channel characteristics.



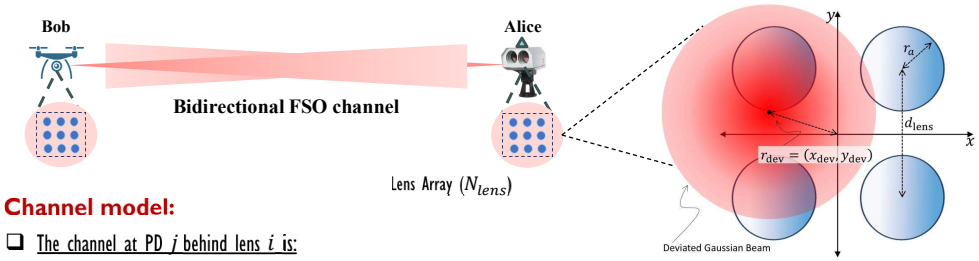
Channel model

- The channel at PD j behind lens i is:

Atmospheric attenuation | Transmitted power | Pointing error | Arrival angle | Turbulence | Noise

$$y_{i,j} = P_t.h_0.\, h_{p,i}(\theta_{ex},\theta_{ey})\; h_{AoA,i,j}(\theta_{AoA,x},\theta_{AoA,y})\; h_{a,i} + n_{i,j}$$

2. Tracking phase – Channel probing and Data measurement



Channel model:

- The channel at PD j behind lens i is:

Pointing error | Arrival angle

$$y_{i,j} = P_t.h_0.\, h_{p,i}(\theta_{ex},\theta_{ey})\; h_{AoA,i,j}(\theta_{AoA,x},\theta_{AoA,y})\; h_{a,i} + n_{i,j}$$

- The pointing error is caused by imperfect tracking: $\theta_e = (\theta_{ex},\theta_{ey})$ (*pointing error angle*)

→ Consequently, the center of the Gaussian beam at the receiver plane is shifted by: $r_{dev}(t) = \begin{bmatrix} x_{dev}(t) \\ y_{dev}(t) \end{bmatrix} = Z_L.\theta_e(t)$

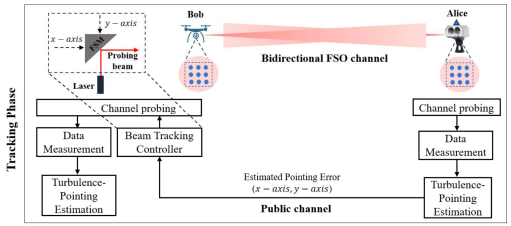
- The instantaneous angle of arrival (AoA) at Alice: $\theta_{AoA} = (\theta_{AoA,x},\theta_{AoA,y}) = \theta_e + \theta_r$

Where $\theta_r = (\theta_{rx},\theta_{ry})$: The received-induced angular jitter

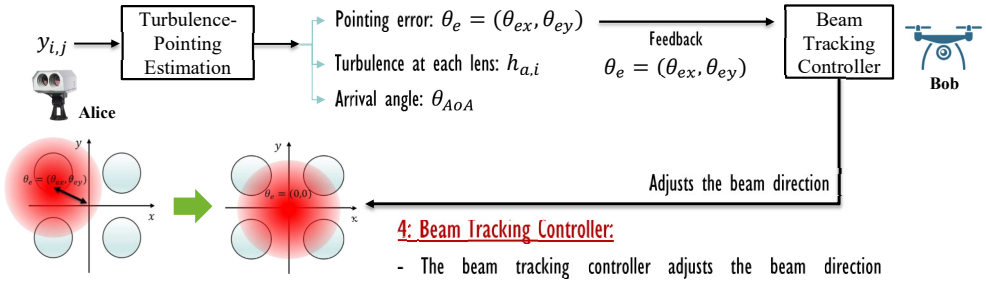
2. Tracking phase – Turbulence-Pointing Estimation

3. Turbulence-Pointing Estimation:

- Based on the received signals, atmospheric turbulence and pointing errors are estimated for beam tracking control and SKG.
- *(More detail about Turbulence-pointing estimation method, please refer to slide 23, 24, 25)*



- The signal at PD j behind lens i is: $y_{i,j} = P_t.h_0.\, h_{p,i}(\theta_{ex},\theta_{ey})\; h_{AoA,i,j}(\theta_{AoA,x},\theta_{AoA,y})\; h_{a,i} + n_{i,j}$



Beam footprint at Alice

4: Beam Tracking Controller:

- The beam tracking controller adjusts the beam direction $\theta_{residual} = -(\theta_{ex},\theta_{ey})$ by using Fast steering mirror (FSM).

System Model – 3. Key generation phase

Key generation phase - Goal

→ Alice and Bob extract the randomness from estimated turbulence to generate secret keys

Key generation phase

- Step 1. Quantization: The estimated turbulence values are converted into binary bits (0 and 1)
- Step 2. Key Reconciliation: Use ECC to correct errors and achieve identical keys
- Step 3. Privacy Amplification: The key is refined to remove any information an eavesdropper might have

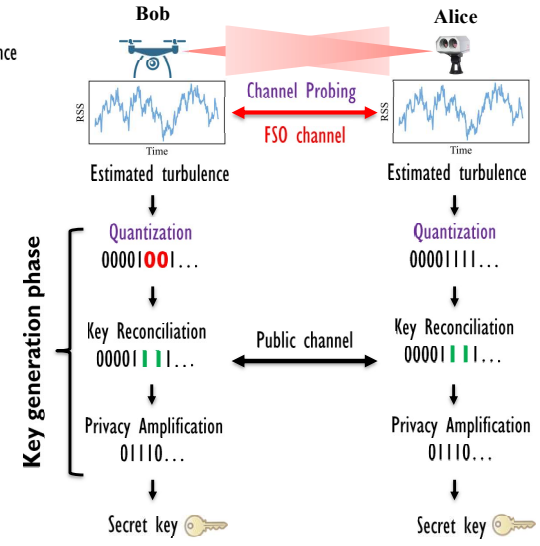


Fig. 6: Key generation process

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Numerical Results and Discussions

Assume: UAV knows the coarse location of BS via GPS (GPS error ≈ 5 m)

TABLE I: System Parameters

Parameters	Values
Sensing phase	
Link length	2000 m
Scanning range	10 m x 10 m
Horizontal step	5 cm
Vertical step	Depen on sensing time
Tracking and Key generation phase	
Beam divergence angle	0.25 mrad
Lens radius	5 cm
Number of lenses	16
Atmospheric turbulence coherence time	1ms
Pointing error coherence time	10ms

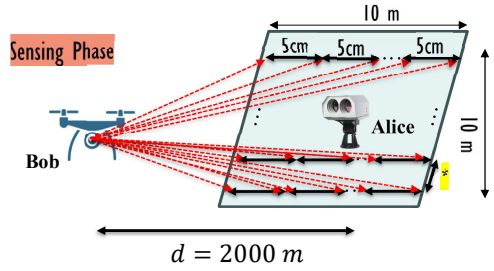


Fig. 7: Illustration of system parameters

Numerical Results and Discussions

❑ Impact of AI tracking (proposed scheme) on pointing error

- Assumption:

  + Channel coherent time of pointing error is 10ms

  + FSM can immediately adjust beam tracking without delay (FSM adaptation delay ~1ms; propagation delay; inference time of ML model ~1ms); FSM has mechanical steering range of approximately 10 mrad along both X-axis and Y-axis

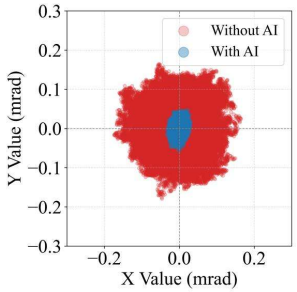


Fig. 8: Pointing error spatial distribution with and without AI tracking

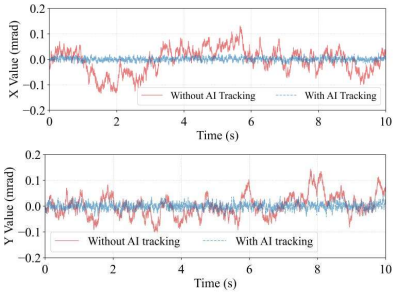


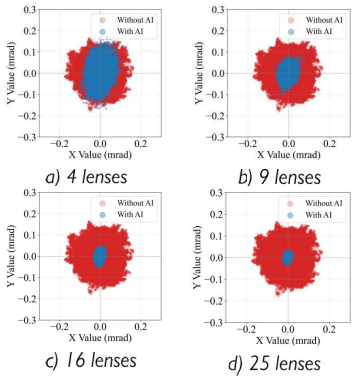
Fig. 9: Time-series performance of pointing error with and without AI tracking

→ The AI-assisted tracking mechanism significantly narrows the spatial fluctuation of pointing errors

→ Key Insights: The proposed design significantly mitigates pointing errors via FSM feedback control

Numerical Results and Discussions

❑ Impact of AI tracking on SKR performance



a) 4 lenses *b) 9 lenses*

c) 16 lenses *d) 25 lenses*

Fig. 10: Pointing error spatial distribution across number of lenses

→ Key insights:

- Increasing number of lenses → narrows the spatial fluctuation of pointing errors → reduce error of turbulence-pointing estimation

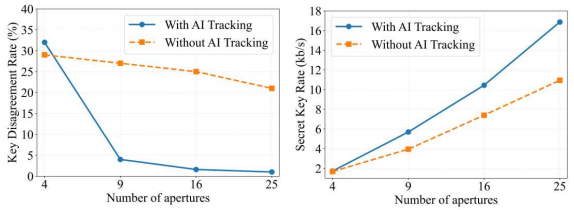


Fig. 11: Impact of number of lenses on KDR and SKR

→ Key insights:

- Increase number of lenses → reduce pointing error → decrease KDR (key disagreement rate)
- Increase number of lenses → enhance source of randomness → increase SKR
- Tradeoff: Increase number of lenses → increase weight for UAV & power consumption

Conclusions

- Contribution:
  - (1) Design sensing and beam tracking algorithm to mitigate pointing errors for lightweight UAVs
  - (2) Improve the SKR by exploiting spatial diversity in multi-aperture FSO system

- Future work:
  - (1) Consider impact of delay (FSM respond time, propagation delay and inference time of ML model) on tracking phase.

Thank you for your attention!

Appendix Slides

2. Tracking phase – Turbulence Pointing Estimation (DETAIL)

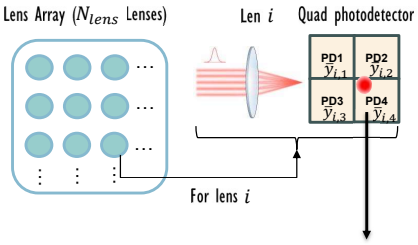
3. Turbulence Pointing Estimation:

Stage I: Noise reduction

Stage II: AoA estimation

Stage III: Pointing error estimation

Stage IV: Jitter + Turbulence estimation



$$y_{i,j} = P_t . h_0 . h_{p,i}(\theta_{ex}, \theta_{ey})\, h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y})\, h_{a,i} + n_{i,j}$$

Stage I: Noise reduction

- Average L_b samples within each channel coherence time: $\bar{y}_{i,j} = \frac{1}{L_b}\sum_{l=1}^{L_b} y_{i,j}[l]$

→ Averaging reduces the noise to: $\bar{n}_{i,j} \sim \mathcal{N}(0, \frac{\sigma_n^2}{L_b})$; the effective noise variance is reduced by $1/L_b$

After Stage I: The received signal can be approximated as noise-free.

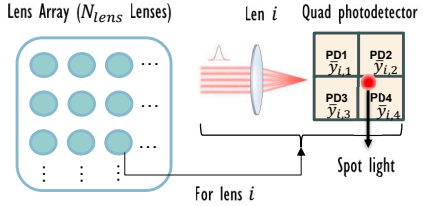
$$\bar{y}_{i,j} \approx P_t . h_0 .\ h_{p,i}(\theta_{ex}, \theta_{ey})\, h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y})\, h_{a,i}$$ → Input step II: AoA estimation

2. Tracking phase – Turbulence Pointing Estimation (DETAIL)

Stage II: AoA estimation $(\theta_{AoA,x}, \theta_{AoA,y})$

- The signal at PD j behind lens i is

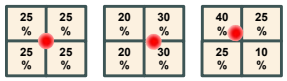
$$\bar{y}_{i,j} = P_t . h_0 . h_{p,i}(\theta_{ex}, \theta_{ey})\, h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y})\, h_{a,i}$$

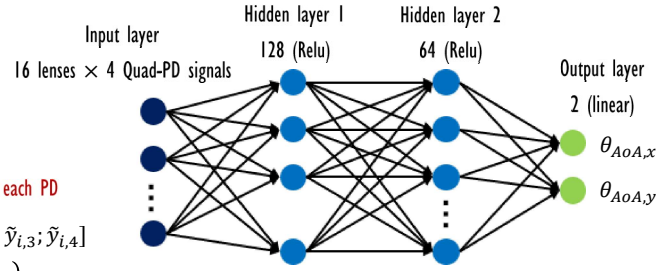


Step1: Removes common scaling

$$\tilde{y}_{i,j} = \frac{\bar{y}_{i,j}}{\Sigma_{k=1}^{4} \bar{y}_{i,k}} = \frac{P_t.h_0.\, h_{p,i}(\theta_{ex},\theta_{ey})\, h_{a,i} h_{AoA,i,j}(\theta_{AoA,x},\theta_{AoA,y})}{P_t.h_0.\, h_{p,i}(\theta_{ex},\theta_{ey}) h_{a,i} \Sigma_{k=1}^{4} h_{AoA,i,j}(\theta_{AoA,x},\theta_{AoA,y})} = \frac{h_{AoA,i,j}(\theta_{AoA,x},\theta_{AoA,y})}{\Sigma_{k=1}^{4} h_{AoA,i,j}(\theta_{AoA,x},\theta_{AoA,y})}$$

Step II: AoA estimation





Key idea: Based on the difference of power in each PD

Input: $\{\tilde{\mathbf{y}}_i\}_{i=1}^{N_{lens}}$ where $\tilde{\mathbf{y}}_i = [\tilde{y}_{i,1}; \tilde{y}_{i,2}; \tilde{y}_{i,3}; \tilde{y}_{i,4}]$

MLP learn to estimate $AoA(\theta_{AoA,x}, \theta_{AoA,y})$

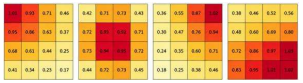
Fig.12: MLP structure for AoA estimation

2. Tracking phase – Turbulence Pointing Estimation (DETAIL)

Stage III: - Pointing Estimation $(\theta_{ex}, \theta_{ey})$

The signal at lens is: $Z_i = h_{p,i}(\theta_{ex}, \theta_{ey})\, h_{a,i}$ (after AoA Compensation)

- Key idea: Pointing estimation is based on the spatial Gaussian distribution.



Example of different pointing profile

- Input: $\{Z_i\}_{i=1}^{N_{lens}}$
- MLP learn to estimate θ_{ex}, θ_{ey}

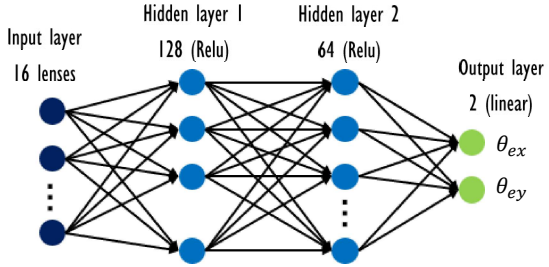


Fig.13: MLP structure for Pointing error estimation

Stage IV: - Turbulence estimation

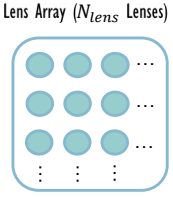
$$h_{a,i} = \frac{Z_i}{h_{p,i}(\theta_{ex}, \theta_{ey})}$$

System Model – 3. Key generation phase

❑ Physical FSO Spatial Correlation Modeling

- Consider an optical aperture array receiver consisting of N lenses
- Spatial irradiance correlation coefficient $\gamma(d_{ij})$ [1] between the $i-th$ and $j-th$ apertures separated by a transverse distance d_{ij} is defined as:

$$\gamma(d_{ij}) = \frac{\zeta(d_{ij})}{\sqrt{\sigma_i^2 \sigma_j^2}}$$



Where:

+ $\sigma_i^2; \sigma_j^2$: the scintillation index of each component receiver
+ $\zeta(d_{ij})$: spatial covariance function of irradiance

→ Based [2] → Full receiver spatial correlation matrix R

$$\mathbf{R} = \begin{bmatrix} 1 & \gamma(d_{12}) & \cdots & \gamma(d_{1N}) \\ \gamma(d_{21}) & 1 & \cdots & \gamma(d_{2N}) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(d_{N1}) & \gamma(d_{N2}) & \cdots & 1 \end{bmatrix}$$

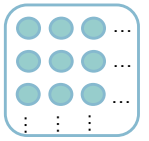
[1] Z. Chen, S. Yu, T. Wang, G. Wu, S. Wang, and W. Gu, "Channel correlation in aperture receiver diversity systems for free-space optical communication," Journal of Optics, vol. 14, no. 12, p. 125710, 2012.
[2] S. M. Navidpour, M. Uysal, and M. Kavehrad, "BER performance of free-space optical transmission with spatial diversity," IEEE Transactions on Wireless Communications, vol. 6, no. 8, pp. 2813–2819, Aug. 2007.

System Model – 3. Key generation phase

❑ Physical FSO Spatial Correlation Modeling

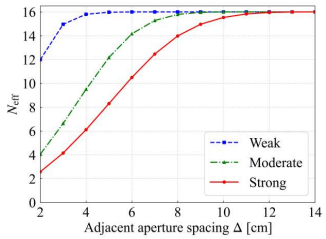
- The receiver spatial correlation matrix R

$$\mathbf{R} = \begin{bmatrix} 1 & \gamma(d_{12}) & \cdots & \gamma(d_{1N}) \\ \gamma(d_{21}) & 1 & \cdots & \gamma(d_{2N}) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(d_{N1}) & \gamma(d_{N2}) & \cdots & 1 \end{bmatrix}$$



- Effective number of independent lenses: $N_{eff} = \frac{N^2}{\Sigma_{i=1}^{N} \Sigma_{j=1}^{N} \gamma(d_{ij})^2}$

→ Key insight:
- When the aperture spacing is large enough, the lenses become independent
- Longer propagation distance increases the correlation among lenses
- Example: At distance 2000m with a 12cm lens spacing, the lenses are independent



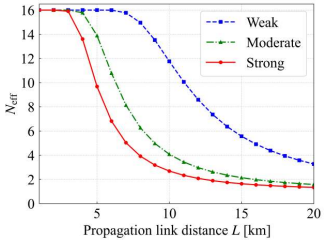


Fig.14: Impact of adjustment aperture spacing and propagation link on Neff